

STEM PLAY LAB

Web Application Upgrade Proposal

Booking, payments, memberships, e-commerce and administration platform

Prepared for: STEM Play Lab

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1. Executive Summary

This proposal covers the transformation of the existing STEM Play Lab website into a complete, responsive web application that supports the organisation's day-to-day operations. The upgraded platform will allow parents and guardians to create accounts, register children, book sessions, make secure payments, purchase memberships and subscriptions, and view their booking history.

The platform will also provide authorised staff with an administration dashboard for managing sessions, capacity, payments, customers, memberships, products, bookings, attendance, enquiries and performance reports. The objective is to reduce manual administration, improve the customer experience and create a scalable digital foundation for the planned Manchester launch and future expansion.

2. Current Platform Assessment

The current prototype already provides a strong public-facing foundation, including:

- Information about STEM classes, build workshops, holiday camps, birthday parties, subscription STEM kits and the smart toy store.
- A four-step session-booking interface with service selection, date selection and parent/child details.
- Interest-registration and contact forms for prospective customers.
- A clear brand message focused on hands-on STEM learning for children aged 3 to 14.
- Responsive navigation and basic accessibility controls.

However, the present version operates mainly as a front-end prototype. Bookings require manual confirmation, no payment is collected during checkout, and there is no customer account area, membership system, database-driven administration dashboard or integrated online store. The proposed upgrade addresses these operational gaps.

3. Project Objectives

- Provide a smooth online journey from discovery to booking and payment.
- Enable customers to manage bookings, children, memberships, subscriptions and purchases from one account.
- Give staff full visibility and control through a secure administration dashboard.
- Automate confirmations, receipts, reminders and important customer communications.
- Support the organisation's classes, workshops, camps, parties, retail products and STEM-kit subscriptions.
- Create a maintainable platform that can support additional locations and services in the future.

4. Detailed Scope of Work

A. Public Website Enhancement

- Refresh and improve the existing pages while retaining the established brand direction.
- Make service information, pricing, age ranges, availability and calls to action easy to manage.
- Improve mobile responsiveness, navigation, form validation, loading performance and accessibility.
- Add legal and operational pages such as privacy policy, terms, refund/cancellation policy and FAQs using client-supplied content.

B. Customer and Family Accounts

- Secure registration, login, logout and password-reset functionality.
- Parent/guardian profile management.
- Multiple child profiles, including age, allergies, medical needs and notes.
- Customer dashboard showing bookings, payment status, memberships, subscriptions, purchases and receipts.

C. Session Booking and Scheduling

- Database-driven setup for classes, workshops, camps, birthday parties and robotics sessions.
- Calendar-based selection of available dates and times.
- Capacity limits, age restrictions, pricing, booking status and waitlist controls.
- Booking confirmation, rescheduling and cancellation workflow subject to business rules.
- Support for single-child and multi-child bookings.

D. Payments and Checkout

- Secure online payment integration using Stripe or another agreed provider available to the client.
- Full or deposit payments for bookings and parties.
- Payment status tracking for successful, pending, failed, refunded and partially refunded transactions.
- Automated payment confirmation and receipt emails.
- Admin view of customers who have paid and those with outstanding balances.

E. Memberships and Subscriptions

- Creation of membership plans with prices, benefits, validity periods and renewal rules.
- One-time or recurring membership payments, depending on the selected payment provider.
- Membership status management: active, pending, expired, cancelled or suspended.
- Subscription STEM-kit plans with delivery frequency and customer subscription history.
- Admin controls for assigning, renewing, cancelling or adjusting memberships.

F. Smart Toy Store and Orders

- Product catalogue for educational toys, robotics kits, STEM games and building sets.
- Product categories, images, descriptions, prices, stock status and featured products.
- Shopping cart, checkout and order confirmation.
- Order-management area for payment status, fulfilment status and customer details.
- Basic inventory quantity management and low-stock visibility.

G. Administration Dashboard

- Secure admin and staff login with role-based access.
- Overview dashboard with bookings, revenue, payments, memberships, customers and order summaries.
- Manage sessions, schedules, capacity, prices, locations and availability.
- View and manage customers, child profiles, enquiries and interest registrations.
- View, filter and export bookings, payments, memberships, subscriptions and orders.
- Record attendance and view basic repeat-booking and customer-retention information.
- Create discount codes or promotional pricing where supported by the agreed payment workflow.

H. Notifications and Communication

- Automated emails for account registration, booking confirmation, payment receipt, cancellation and password reset.
- Session reminders and basic membership/subscription notifications.
- Admin notification when a new booking, payment, order or enquiry is received.
- Editable email templates for key transactional messages.

I. Reporting and Data

- Basic reporting for bookings, revenue, attendance, membership status, orders and popular services.
- Date-range filters and CSV export for selected records.
- Dashboard indicators that support the organisation's measurement-led approach.
- Structured database suitable for future advanced analytics and multi-location reporting.

J. Security, Testing and Launch

- Secure authentication, protected admin routes and safe handling of customer information.
- Server-side validation and basic protection against common web-application vulnerabilities.
- Cross-device and major-browser testing.
- Production deployment, environment configuration and payment-webhook setup.
- Basic administrator handover guide and launch support.

5. Key User Roles

Role	Primary Access	Main Activities
Visitor	Public pages	Review services, register interest, browse sessions and products.
Customer	Customer account	Manage family details, book, pay, join memberships, subscribe and purchase products.
Staff	Limited admin tools	View bookings, customer information, schedules and attendance based on permissions.
Administrator	Full dashboard	Manage the entire platform, users, payments, sessions, memberships, orders and reports.

6. Proposed Technical Approach

The final technology stack will be confirmed before development begins. A suitable modern implementation would include:

Component	Proposed approach
Front end	A responsive React/Next.js interface based on the existing design direction.
Back end and database	A managed application back end and relational database such as Supabase/PostgreSQL or an equivalent agreed solution.
Payments	Stripe Checkout and recurring billing, subject to the client's verified payment-provider account and supported country requirements.
Hosting	Managed deployment through a suitable provider such as Vercel/Netlify, with separate production configuration.
Email	A transactional email provider for confirmations, receipts and reminders.
Administration	A custom, protected dashboard integrated with the same application database.

7. Delivery Plan and Timeline

Phase	Stage	Activities	Target
1	Discovery and setup	Confirm business rules, user flows, technical stack, data structure and payment-provider requirements.	Week 1
2	Core platform	Customer accounts, child profiles, database, services, schedules and booking workflow.	Weeks 1
3	Commerce and memberships	Payments, deposits, memberships, subscriptions, products, cart and orders.	Weeks 2

Phase	Stage	Activities	Target
4	Administration and reporting	Admin dashboard, management tools, notifications, attendance and reports.	Weeks 3
5	QA and launch	Testing, client review, agreed fixes, deployment, handover and launch support.	Weeks 3

Timeline assumption: The client provides content, policy decisions, payment-account access and feedback without material delay. Changes to the agreed scope may require a revised schedule and fee.

8. Deliverables

- Completed production-ready web application with the agreed functionality.
- Responsive customer-facing website and customer account area.
- Secure administration dashboard.
- Booking, payment, membership, subscription and e-commerce workflows.
- Configured database, transactional emails and deployment environment.
- Basic admin handover documentation.
- Fourteen days of post-launch defect support for items within the agreed scope.

9. Items Not Included

- Professional copywriting, photography, video production, logo redesign or large-scale content entry.
- Native iOS or Android mobile applications.
- Custom accounting, payroll, school-management or courier integrations not expressly listed in this proposal.
- Physical fulfilment, shipping operations, tax advice, legal drafting or regulatory certification.
- Major features or business-model changes requested after approval of the agreed requirements.

10. Client Responsibilities and Assumptions

- Provide final service descriptions, prices, schedules, membership rules, policies, product information and legal text.
- Create and verify required third-party accounts, including the payment-provider account.
- Provide timely access to the existing source code, domain/DNS settings and hosting accounts where applicable.
- Nominate one primary contact who can consolidate feedback and approve milestones.
- Review submitted milestones promptly and provide clear, consolidated feedback.
- Confirm that the client has the right to use all supplied text, images, logos and customer information.

11. Revisions, Change Requests and Support

The fixed price includes up to two consolidated revision rounds during the client-review stage, provided the requested changes remain within the agreed scope. New modules, materially different workflows, extensive redesigns or additional integrations will be quoted separately before work proceeds.

The fourteen-day post-launch support period covers correction of reproducible defects in the delivered functionality. It does not cover new features, third-party outages, content changes, misuse of the system or changes made by another developer.

12. Acceptance

Approval of this proposal confirms agreement with the scope, fixed project fee, payment schedule, assumptions and exclusions stated above. A final requirements checklist will be agreed at project commencement and will serve as the reference for delivery and acceptance.

Signatures

For STEM Play Lab	For Adewale Ajayi
Name: _____	Name: Adewale Ajayi
Signature: _____	 Signature: _____
Date: _____	Date: 26 July, 2026

Proposal validity: 14 calendar days from the date above.